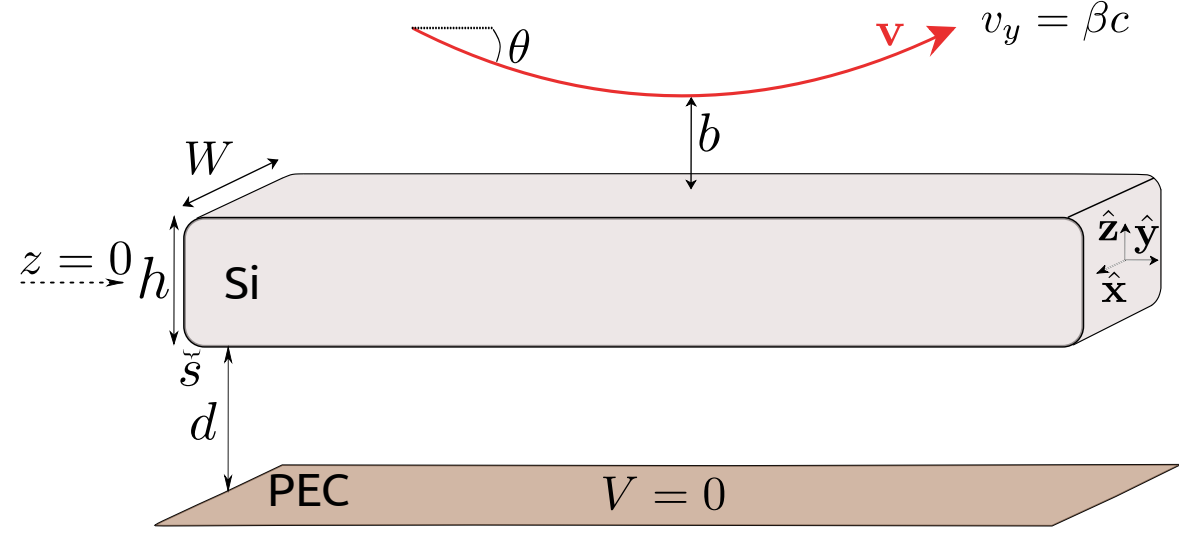
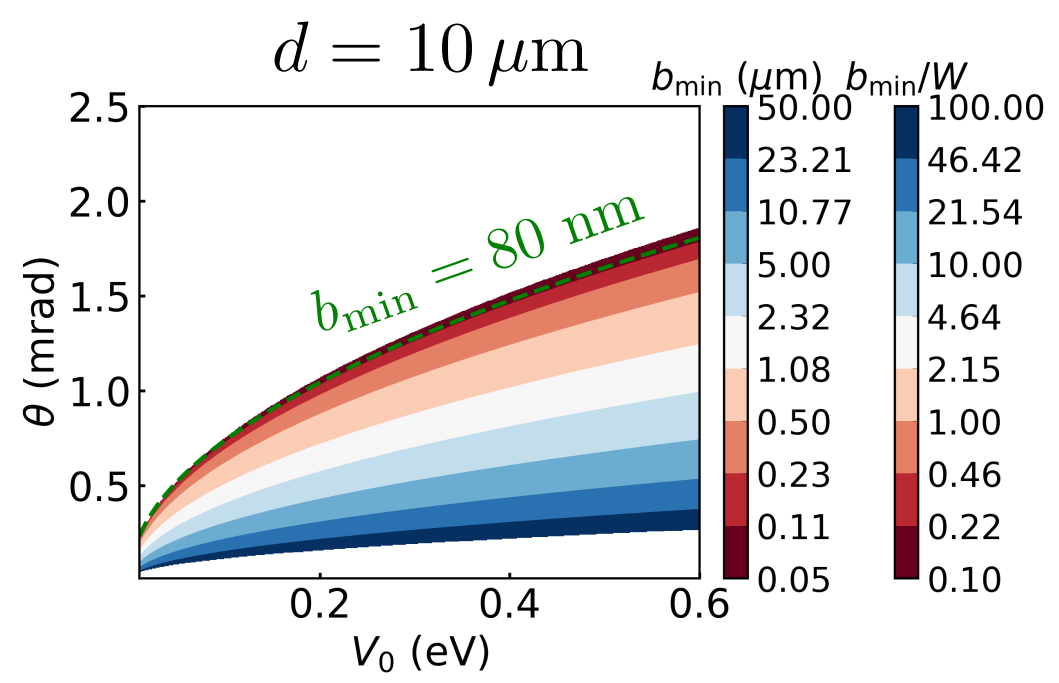
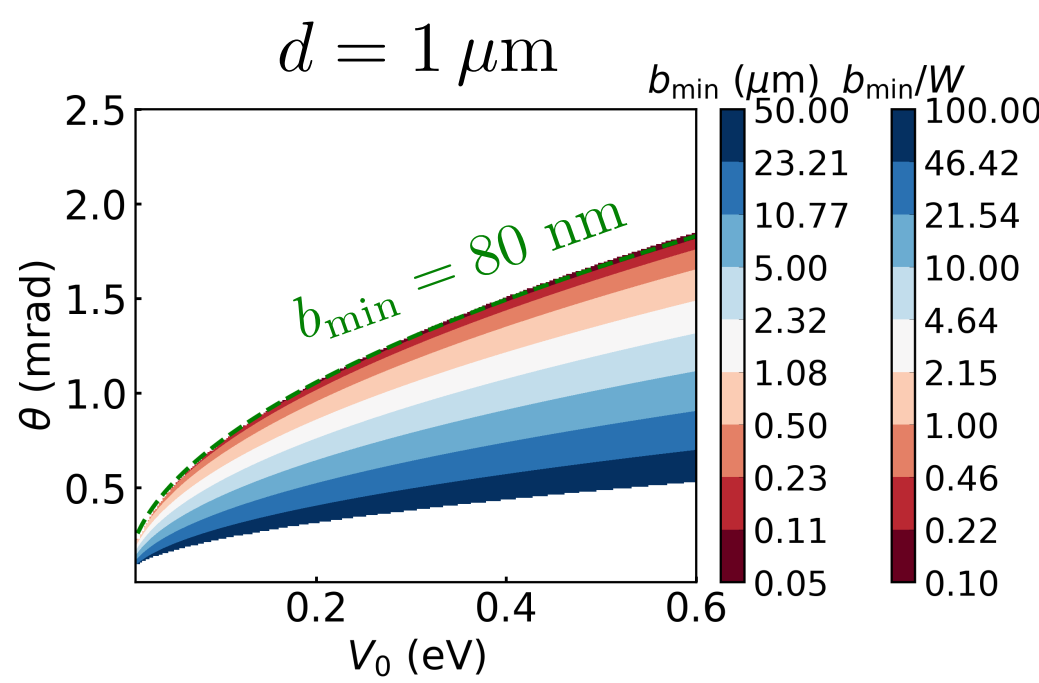


Electron-coupling to waveguide

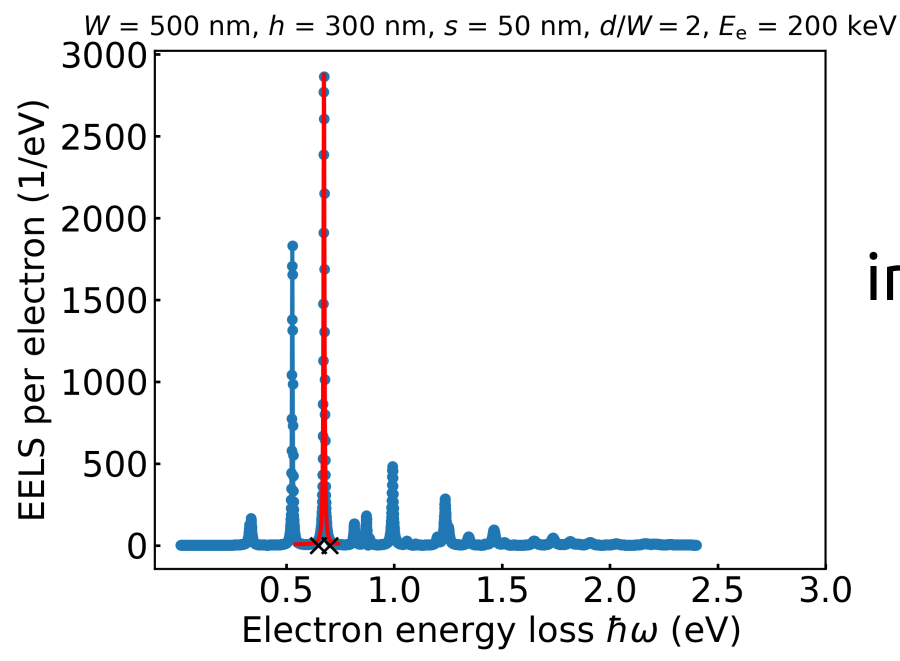


$W = 500 \text{ nm}$ $s/W = 0.1$
 $h = 300 \text{ nm}$ $E_e = 100 \text{ keV}$

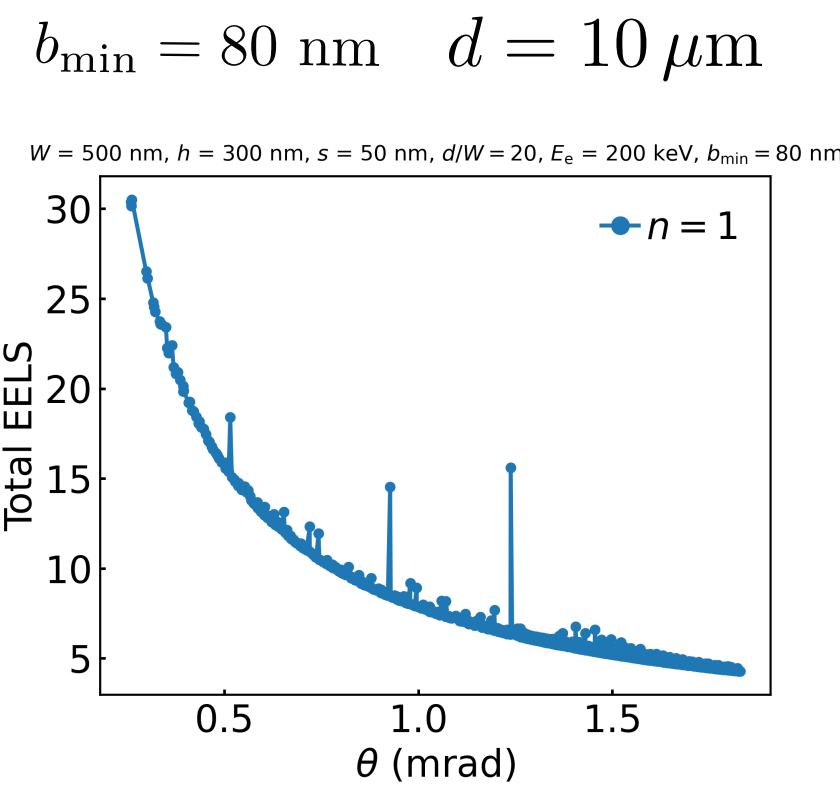
$\Gamma_{\text{EELS}}(\omega) = 2 \int_{z_{\min}}^{\infty} \frac{vdz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, z)}{dy}$

$x_e = 0$

EELS integrated over z:



integration over peak →



200 keV - $\lambda_e = 2.5 \text{ pm}$ 0.8 nm 800 nm
100 keV - $\lambda_e = 3.7 \text{ pm}$ 1.2 nm 1200 nm
 W_0 20
 $\theta = 1 \text{ mrad}$
 $\theta = 0.1$ $\theta = 0.3$ μm
 $V_0 = 1 \text{ meV}$ $\times 10$ $\times 100$ $\times 10^3$

- Running in sonic many frequencies (~7000 frequencies for 1 value of z)
- Writting the codes in c to do everything in sonic faster
- Discussing with Saad